PHILIP ADAM LEMAITRE

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RESEARCH PROFILE

My research lies at the intersection of quantum foundations, relativistic quantum information, and classical/quantum machine learning for scientific discovery. I study how contextuality, observer-dependence, and information-processing tasks behave in spacetime-sensitive settings, with current work on contextuality harvesting from quantum fields, relativistic quantum computing, and interpretable reinforcement-learning agents that can learn physically meaningful structure. In collaboration with Johannes Fankhauser, I am also developing a realist-relational formulation of quantum theory that treats the subject-object relation as a central physical and conceptual feature.

RESEARCH INTERESTS

- Quantum foundations, contextuality, and cohomological approaches to contextuality
- Relativistic quantum information, relativistic quantum computing, and quantum field-mediated information processing
- Emergence and reconstruction of spacetime from algebraic approaches to quantum gravity and information
- Observers, knowledge, and realist-relational formulations of quantum theory
- Interpretable classical and quantum machine learning, especially reinforcement-learning agents that learn spacetime structure
- General frameworks for modelling physical agents and scientific discovery

EDUCATION

University of Innsbruck, Austria. *July 2023 - Present*
Ph.D. Physics
Current doctoral research in relativistic quantum information, contextuality harvesting, and interpretable artificial intelligence
Supervised by Hans J. Briegel

University of Waterloo, Canada. *Sep. 2020 - Oct. 2022*
M.Sc. Physics
Nominated for the Dean of Science award
Thesis: "Atomic Shell Structure in a Ring Polymer Formulation of Orbital-Free Density Functional Theory"
Supervised by Russell B. Thompson

University of Guelph, Canada. *Sep. 2015 - Feb. 2020*
B.Sc., Honours - Theoretical Physics
Recognized on the Dean's Honours List (2017-2020)
Thesis: "An Analysis of the Equilibrium Configurations and Stability of Thin Spherical Matter Shells in Relativistic and Newtonian Gravity"
Supervised by Eric Poisson

PUBLICATIONS

P. A. LeMaitre, "Harvesting Contextuality from the Vacuum". *Phys. Rev. D 113, 045001 (2026).*

P. A. LeMaitre, T. R. Perche, M. Krumm, and H. J. Briegel, "Universal Quantum Computer from Relativistic Motion". *Phys. Rev. Lett. 134, 190601 (2025).*

M. A. Kealey, **P. A. LeMaitre**, and R. B. Thompson, "Fermion exchange in ring polymer quantum theory". *Phys. Rev. A 109, 052819 (2024).*

P. A. LeMaitre, M. Krumm, and H. J. Briegel, "Multi-Excitation Projective Simulation with a Many-Body Physics Inspired Inductive Bias". *Artificial Intelligence 352, 104489 (2026).*

P. A. LeMaitre and R. B. Thompson, "On the Origins of Spontaneous Spherical Symmetry-Breaking in Open-Shell Atoms Through Polymer Self-Consistent Field Theory". *J. Chem. Phys. 158, 064301 (2023).*

P. A. LeMaitre and R. B. Thompson, "Gaussian Basis Functions for a Polymer Self-Consistent Field Theory of Atoms". *Int. J. Quantum Chem.* 123(12), e27111 (2023).

P. LeMaitre and E. Poisson, "Equilibrium and stability of thin spherical shells in Newtonian and relativistic gravity". 87, 961-970 (2019).

MANUSCRIPTS IN PREPARATION

Philip A. LeMaitre, Giacomo Franceschetto, and Hans J. Briegel. "Interpretability-performance tradeoffs in photonic quantum machine learning using wave-particle duality." Manuscript in preparation.

Philip A. LeMaitre, T. Rick Perche, and Hans J. Briegel. "Learned representations of spacetime inside an agent's memory from relativistic quantum information tasks." Manuscript in preparation.

RESEARCH EXPERIENCE

Doctoral Researcher, Institute for Theoretical Physics, University of Innsbruck *July 2023 - Present*
Supervised by Hans J. Briegel

- Developed detector-field protocols for studying how contextual correlations can be extracted from the vacuum state of a quantum field.
- Analyzed relativistic motion as a resource for quantum information processing, including co-authored work on universal quantum computation from relativistic dynamics.
- Investigate contextuality and observer-dependent structure in spacetime-sensitive quantum systems, with an emphasis on foundations, relativistic quantum information, and algebraic perspectives.
- Built and studied interpretable projective-simulation and reinforcement-learning models in which memory structure, exploration, and learned policies can be inspected directly.
- Doctoral research is currently funded through my supervisor's grant.

Master's Thesis, Department of Physics and Astronomy, University of Waterloo *Sep. 2020 - Aug. 2022*
Supervised by Russell B. Thompson

- Developed a classical self-consistent field-theoretic model of ring polymers in 3+1 dimensions equivalent to quantum density functional theory in 3 dimensions.
- Predicted spontaneous shell structure and symmetry-breaking in neutral atoms up to neon without the use of atomic orbitals.
- Extended the method across the periodic table using partial shell information and spherical averaging, obtaining binding energies within 3% of Hartree-Fock values up to krypton and within 10% up to radon.
- Developed a spectral formulation for general non-orthogonal basis functions and designed a numerical solver using angular Gaussian basis functions. (Access thesis here)

Bachelor's Thesis, Department of Physics, University of Guelph *Sep. 2018 - June 2019*
Supervised by Eric Poisson

- Developed a simplified stellar model of compact stars based on thin spherical shells in both relativistic and Newtonian gravity.
- Derived equilibrium sequences and conditions connecting the vanishing of the mass derivative with the onset of dynamical instability.
- Identified stable oscillatory and unstable collapse regimes and implemented a numerical integration routine for the full shell dynamics.
- Used the model to illustrate the link between the maximum mass of a neutron star and gravitational collapse using methods accessible to introductory students of general relativity.

FUNDING AND AWARDS

Doctoral Research Support, University of Innsbruck *July 2023 - Present*
Doctoral research supported through my supervisor's grant.

IODE Gladys Raiter Bursary for Graduate Study *Oct. 2021*
Awarded to graduate students with ties to Waterloo region and with high overall academic standing in their program. Valued at 5000 CAD.

Science Graduate Award *Sep. 2020 - Aug. 2022*
Awarded to full-time thesis-based graduate students at the University of Waterloo. Valued for at least 1500 CAD per term.

Marie Curie Graduate Student Award *Sep. 2020 - Aug. 2022*
Awarded for academic excellence during graduate study at the University of Waterloo. Valued for at least 800 CAD per term.

Graduate Research Studentship *Sep. 2020 - Aug. 2022*
Awarded to aid thesis-based graduate students at the University of Waterloo in the completion of their thesis. Valued for at least 2285 CAD per term.

Linda S. Allen Memorial Graduation Prize *June 2020*
Awarded for overcoming significant obstacles and remaining in good academic standing during the completion of a bachelors degree at the University of Guelph. Valued at 1000 CAD.

Eglestaff Scholarship *Mar. 2019*
Awarded for academic excellence during the third year of undergraduate study at the University of Guelph. Valued at 750 CAD.

PRESENTATIONS

Invited Talks

LeMaitre, Philip A.. "Harvesting Contextuality from the Vacuum". *May 2026*
Quantum Information Group Seminar, Leibniz University, Hannover

LeMaitre, Philip A.. "A Universal Quantum Computer From Relativistic Motion". *Mar. 2026*
Quantum Foundations Group Seminar, NORDITA, Stockholm

Contributed Talks

LeMaitre, Philip A.. "A Multi-Excitation Projective Simulation Agent". *Aug. 2026*
IJCAI 2026, Bremen, Germany

LeMaitre, Philip A.. "Harvesting Contextuality from the Vacuum". *June 2026*
RQI Circuit 2026, Wien, Austria

LeMaitre, Philip A.. "A Universal Quantum Computer From Relativistic Motion". *June 2025*
15th Relativistic Quantum Information North Conference, University of Naples Federico II, Napoli, Italy

LeMaitre, Philip A.. "Physical Characteristics and Manifestations of Agents". *Sep. 2024*
Innsbruck-Konstanz-Hannover Meeting on Physics and Philosophy, Maria Waldrast, Austria

LeMaitre, Philip A.. "A Universal Quantum Computer From Relativistic Motion". *Aug. 2024*
14th Relativistic Quantum Information North Conference, Charles University, Prague, Czechia

LeMaitre, Philip A.. "A Multi-Excitation Projective Simulation Agent". *Mar. 2024*
DPG Spring Meeting 2024, TU Berlin, Berlin, Germany

LeMaitre, Philip A.. "Spontaneous Spherical Symmetry-Breaking in Open-Shell Atoms Through Polymer Self-Consistent Field Theory". *Nov. 2022*
36th Symposium on Chemical Physics, University of Waterloo, Waterloo, Canada

LeMaitre, Philip A.. "Atomic Shell Structure in a Polymer-Based Approach to Orbital-Free Density-Functional Theory". *June 2022*
2022 Canadian Association of Physicists Congress, McMaster University, Hamilton, Canada

LeMaitre, Philip A.. "Thin-Shells as Neutron Stars: A Simple Stellar Model With Great Insight". *Nov. 2019*
Canadian Undergraduate Physics Conference 2019, McGill University, Montreal, Canada

Posters

LeMaitre, Philip A.. "Harvesting Contextuality from the Vacuum". *April 2026*
Observers and Causality in Quantum Gravity Conference, Bratislava, Slovakia

LeMaitre, Philip A.. "A Universal Quantum Computer from Relativistic Motion". *April 2025*
QISS 2025 Conference, University of Vienna, Vienna, Austria

LeMaitre, Philip A.. "Quantum Multi-Excitation Projective Simulation". *June 2024*
6th Seefeld Quantum Information Workshop, Wellnesshotel Schönruh, Seefeld, Austria

LeMaitre, Philip A.. "Towards a Multi-Excitation Projective Simulation Agent". *Sep. 2023*
Innsbruck-Konstanz Meeting on Physics and Philosophy, Lamm Hotel, Bregenz, Austria

MEDIA APPEARANCE

New Scientist *April 2025*
Interviewed by Karmela Padavic-Callaghan for coverage of "A Universal Quantum Computer from Relativistic Motion": "We can build quantum computers using the rules of special relativity."

TEACHING EXPERIENCE

Mathematics and Physics Tutor, Paper Tutoring *Sep. 2022 - Mar. 2023*

- Tutored students from elementary to high school level in mathematics and physics, including lesson planning, exercise preparation, progress reporting, and communication with prospective clients.

Teaching Assistant, University of Waterloo *Sep. 2020 - Aug. 2022*

- PHYS 111 - Physics I: proctoring examinations, supervising student groups, leading tutorials, grading assignments, moderating the student question-and-answer forum via Discord.
- NE 216 - Advanced Calculus and Numerical Methods I: grading tests and holding office hours.
- PHYS 233 - Introduction to Quantum Mechanics: supervising student groups, leading tutorials, holding office hours, and grading assignments and progress reports.
- PHYS 256L - Optics Laboratory: grading lab reports and final projects, virtual office hours via Piazza.
- PHYS 442 - Electricity and Magnetism III: grading assignments.

Mathematics Tutor, University of Guelph *Sep. 2019 - June 2020*

- Tutored undergraduate students in linear algebra, calculus, advanced calculus, and differential equations, including planning and preparing problem sets.

SERVICE, OUTREACH, AND MENTORING

Co-organizer of Public Outreach Events, Briegel Group, University of Innsbruck *2024 - Present*

- Co-organize the design and delivery of public-facing outreach activities that introduce students to artificial intelligence, reinforcement learning, and theoretical physics.
- Helped develop an interactive projective-simulation reinforcement-learning platform where visitors vary agent and environment parameters and watch a learned policy form in real time.
- Helped build a modular group research poster and adaptable booth format, and present live demonstrations translating exploration, reward shaping, memory, and learned policies into accessible activities.

Master's Student Mentoring, University of Innsbruck

- Provided guidance to a master's student on research orientation, conceptual background, and early project direction.

Peer Mentoring, University of Guelph and University of Waterloo *Sep. 2017 - Sep. 2022*

- Assisted fellow students with course material through explanation and problem-solving guidance.
- Offered physics-related career advice and directed students toward helpful academic resources.

SOFTWARE AND OPEN-SOURCE CONTRIBUTIONS

Open-Source Contributor, SymPy *Sep. 2022 - Present*

- Contributed symbolic functionality to the physics submodule of SymPy for computing integrals of three real spherical harmonics, extending the package's support for angular-momentum and basis-function calculations.
- Develop additional symbolic quantum-physics utilities for the package.

Research Code Development and Reproducibility *2020 - Present*

- Maintain research repositories for contextuality harvesting, relativistic quantum computing, polymer self-consistent field theory, and numerical physics projects, including code used to reproduce calculations, generate figures, and test model behaviour.
- Develop custom numerical workflows for detector-field models, self-consistent field calculations, reinforcement-learning agents, and symbolic or spectral calculations.

PROFESSIONAL EXPERIENCE

Expert AI Data Trainer in Mathematics, Invisible Technologies *Oct. 2022 - Sep. 2023*

- Used a combination of knowledge on pedagogy and advanced mathematics to evaluate and train natural language-based artificial intelligence engines to solve mathematical problems
- Proactively outlined new research directions for the client and created process documentation

- Helped to onboard/mentor new team members, creating demonstration videos and workflow documentation for them
- Helped create quality control processes for the team and other prospective clients, along with metrics to benchmark model performance
- Worked collaboratively with a small team of experts whose work aided the client in launching one of the biggest public artificial intelligence products

SKILLS

Research Skills

- Quantum information theory, including contextuality, relativistic quantum information, quantum computing, and quantum-field-mediated information-processing protocols
- Quantum foundations and conceptual analysis of observers, knowledge, operational structure, and subject-object relations in quantum theory
- Analytical modelling for detector-field systems, spacetime-sensitive quantum protocols, compact-object toy models, and self-consistent field theories
- Numerical simulation and spectral methods for quantum information, quantum chemistry, reinforcement learning, and differential-equation-based physical models
- Interpretable machine learning and reinforcement learning, with emphasis on projective simulation, memory networks, policy visualization, and physically structured agents
- Scientific writing, technical presentation, outreach communication, and cross-field synthesis across physics, computation, and philosophy

Computing Skills

- Programming and tooling: Python, LaTeX, Git/GitHub, Fortran
- Scientific workflows: reproducible numerical experiments, symbolic manipulation, data analysis, figure generation, and custom research-code development
- Operating systems: macOS, Windows, Ubuntu Linux

AFFILIATIONS

- Member of the International Society for Relativistic Quantum Information *Feb. 2026 - Present*
- Member of the RQI COST Action *Jan. 2026 - Present*
- Regular member of the Canadian Association of Physicists *Oct. 2022 - Oct. 2023*
- Graduate member of the Canadian Association of Physicists *Nov. 2020 - Oct. 2022*
- Undergraduate affiliate member of the Canadian Association of Physicists *Nov. 2019 - Nov. 2020*